Data & AI

17. The Amount of Information

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Today

- Assignment 3
- Final project and its schedule
- The amount of information

Assignment 3

Assignment 3

A question that a language model answers wrongly

- A person with the right experience knows the answer
- The model looks able to answer it
- DeepSeek V4.1 Flash with web search gets it wrong or leaves something out
- Due Sunday 29 November, 22:00

Where to Look

- Simple-looking tasks: reading a clock, counting letters (lecture 16)
- Knowledge that people with experience share but do not write on the web
  - How a task is really done at a part-time job, in a club, in a hobby
  - Customs that people do not state in public
- Things an experienced person checks and the model leaves out
  - The other person's situation, the order of steps, what goes wrong in practice

Questions That Do Not Count

- "What is my dog's name?": nobody around you could know it either
- "Prove the Riemann hypothesis": experts cannot answer it either
- A question whose answer nobody can check

What to Submit

- The question
- The model's answer
- What a person with experience knows or checks
- What is wrong or missing in the answer

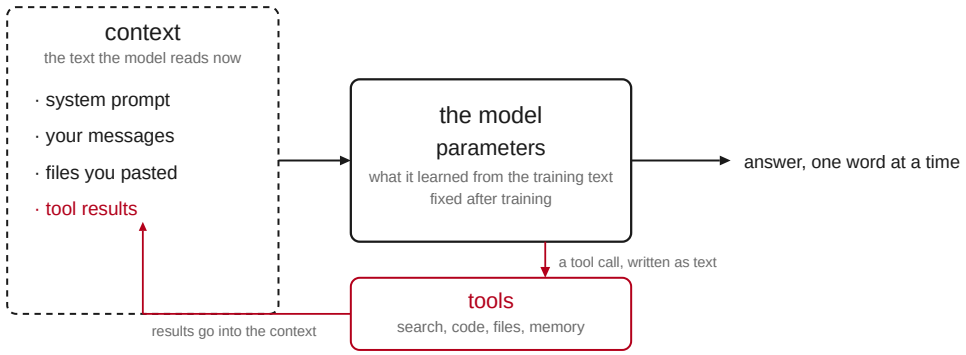
Trying a Stronger Model

- I will set up Claude for the class
- Each student can send a few questions and read the answers
- It takes some time to prepare

Final Project

What a Coding Agent Does Not Have

Lecture 15: an answer comes from the parameters, the context, or a tool



- A difficulty in your own daily life
- What bothers you in an app you use
- A game you want to play
- Why making it is enjoyable for you

The Purpose of the Final Project

You
decide what to make
and why you want it

→

Coding agent
implements it

→

An app
that you use

- The project is practice for the first step

Final Project

An app or a tool that you, or people around you, would use

- Built with a coding agent
- Useful or just fun: a game is fine
- It fits your taste and your situation better than what already exists
- A model inside: an image model, a small language model, or any LLM API
- No AI at all: accepted if you really want it and it clearly differs from what exists

The Proposal

One slide, two minutes

- What it does
- Who uses it, and when
- How it differs from what already exists
- Which model it includes

Help with the Design

- Write what you want in specific detail
- I will make an implementation design with Claude, a few times for each student
- Give the design to your coding agent

Schedule

Date	
Mon 23 Nov, 22:00	Proposal slide due
Tue 24 Nov, Thu 26 Nov	Proposal presentations
Sun 29 Nov, 22:00	Assignment 3 due
Tue 8 Dec, Thu 10 Dec	Project work in class
Tue 15 Dec, Thu 17 Dec	Final presentations

The Amount of Information

The Up-Down Game

A number from 1 to 50 is printed inside a soju bottle cap

- One person looks at the number
- The others call a number in turn
- The answer is "up", "down", or "correct"

Possible Cases

Before any answer: 50 possible numbers, and one of them is true

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

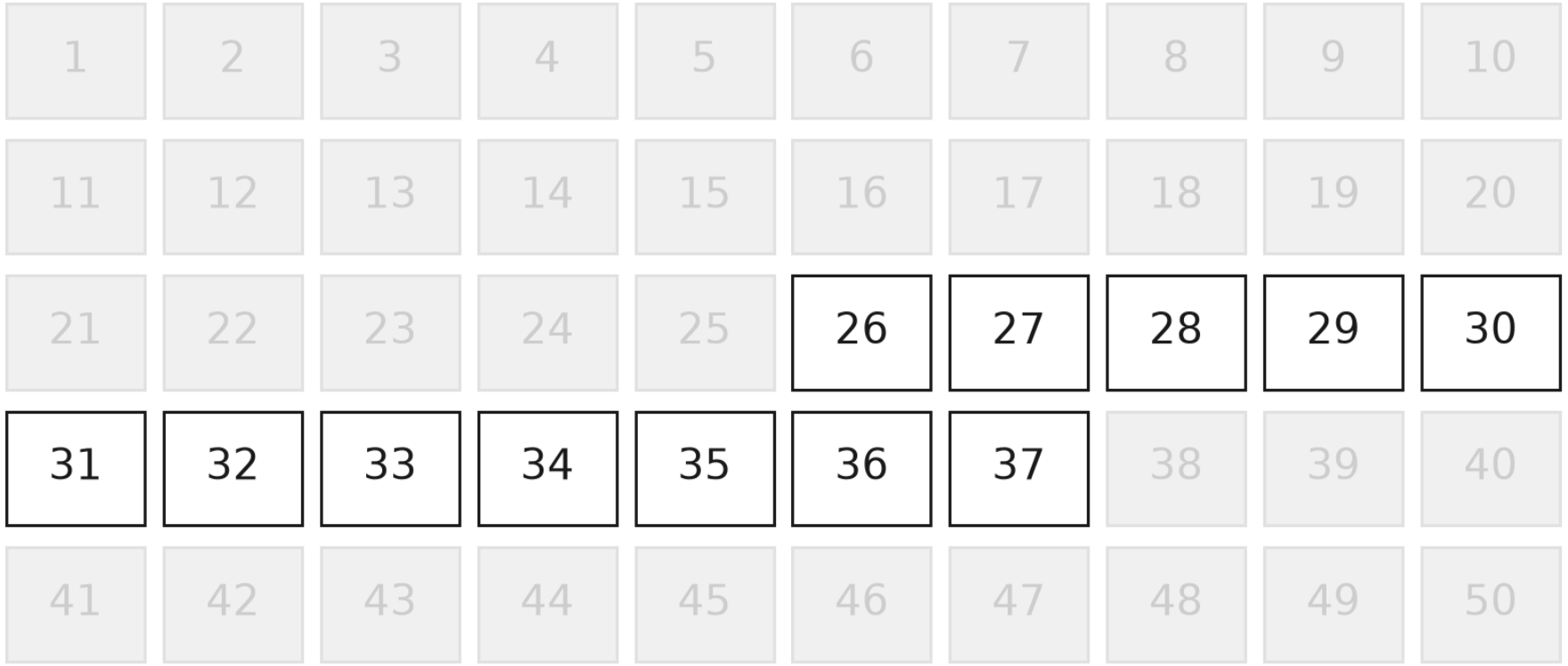
Possible Cases

Call 25, "up": 25 possible numbers

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

Possible Cases

Call 38, "down": 12 possible numbers



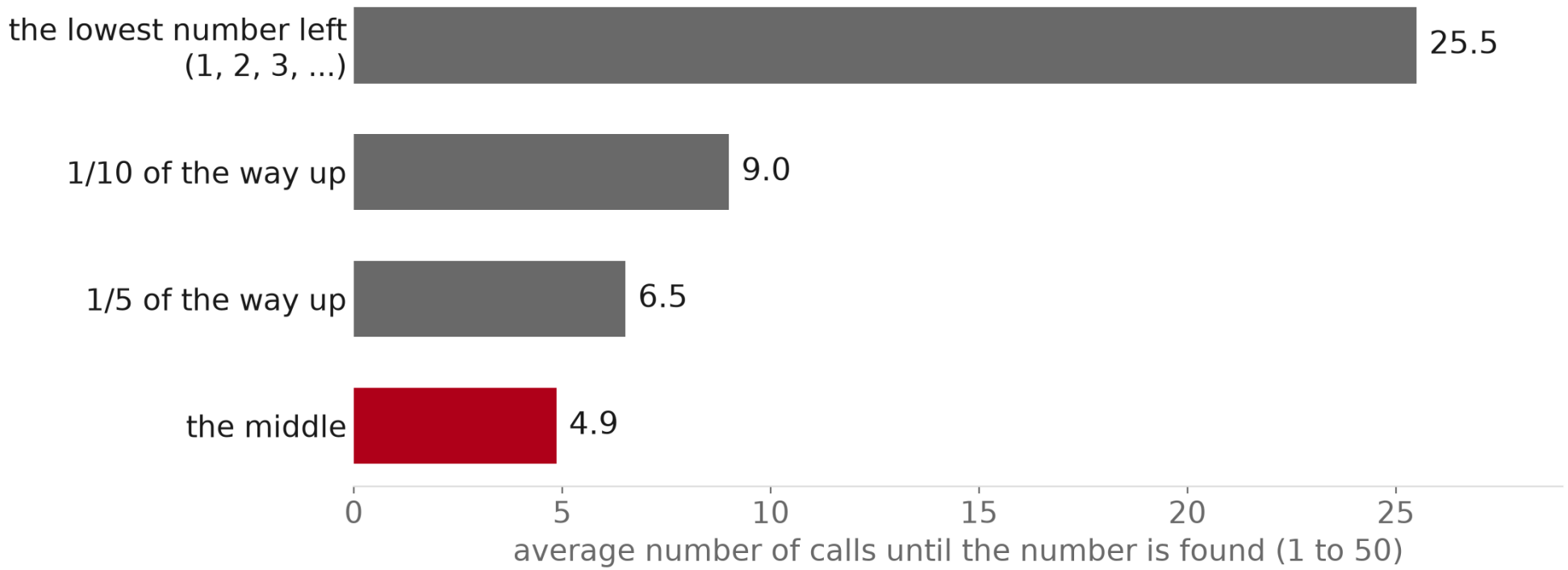
Possible Cases

With all the information: one number, the true one

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	**37**	38	39	40
41	42	43	44	45	46	47	48	49	50

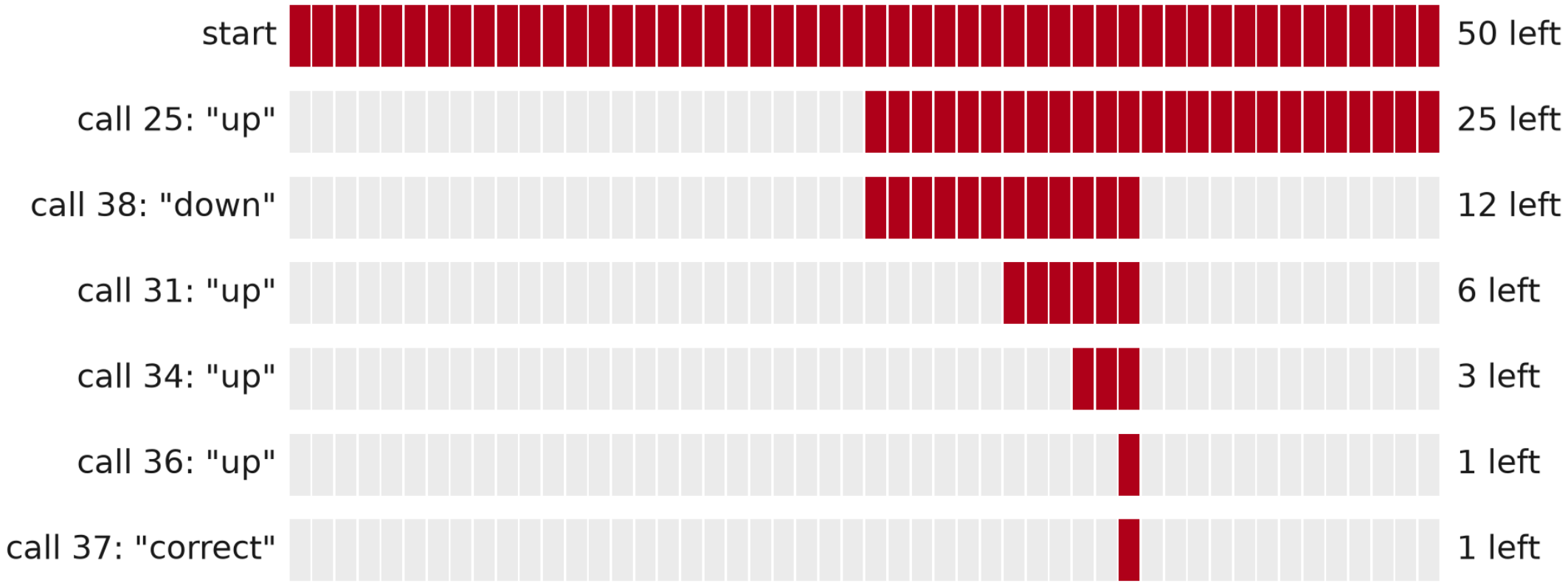
Four Ways to Choose the Next Number

The average over all 50 possible numbers



One Game with the Middle

The number in the cap is 37



- Each answer leaves about half of the numbers

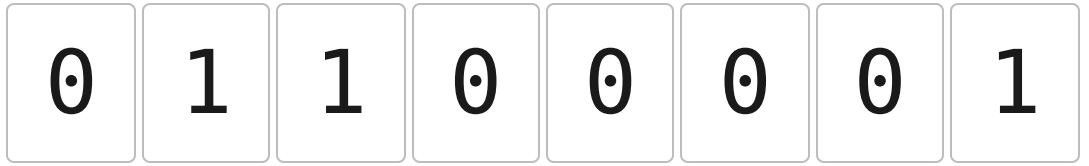
Bit

One of two values, written as 0 or 1

0	1
no	yes
tails	heads
off	on

Byte

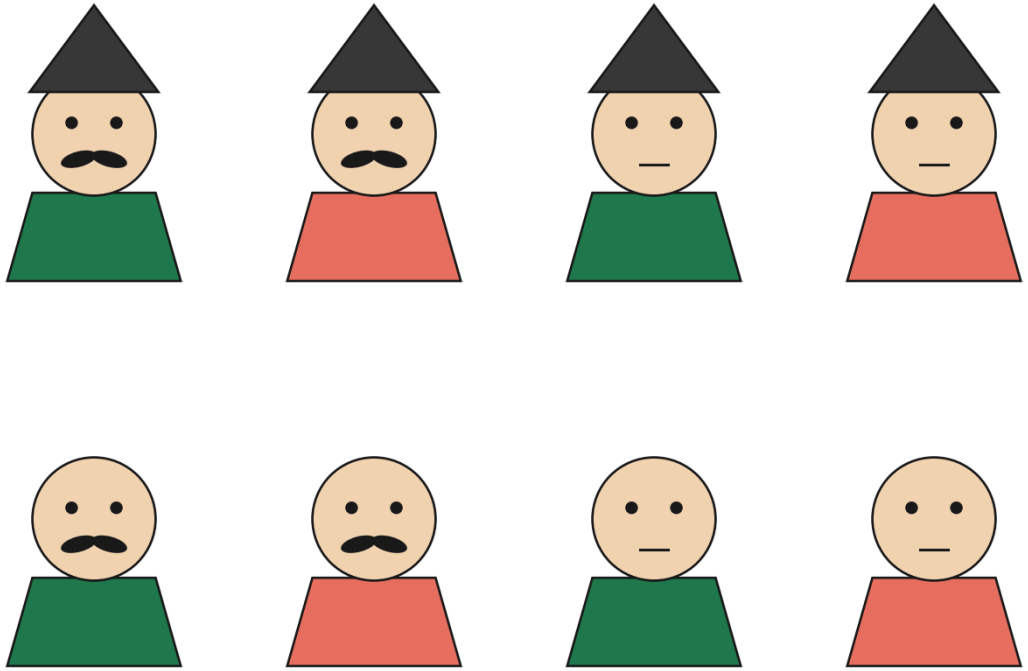
Eight bits



the letter "a"

Eight Suspects

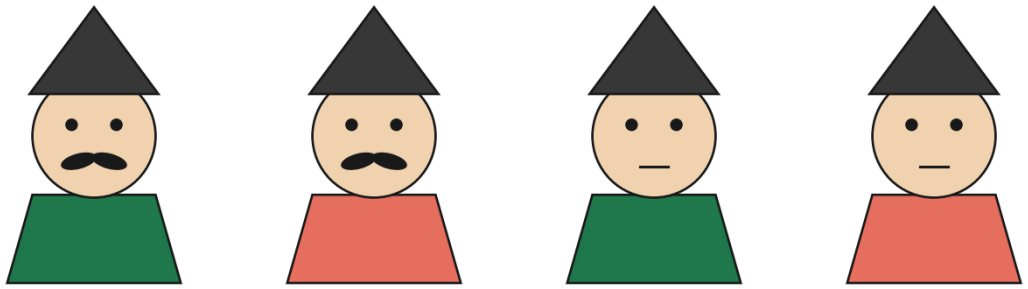
A witness saw one of them



- A hat, or no hat
- A mustache, or no mustache
- A green shirt, or a red shirt
- Every combination appears once

Eight Suspects

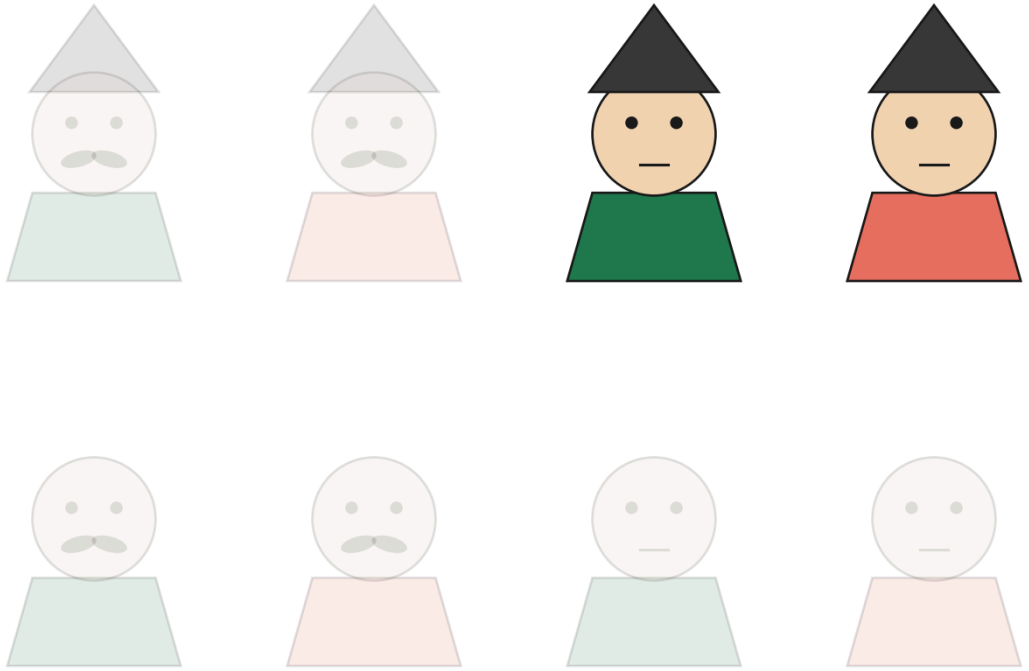
A witness saw one of them



Question	Answer	Bit	Left
			8
Wearing a hat?	yes	1	4

Eight Suspects

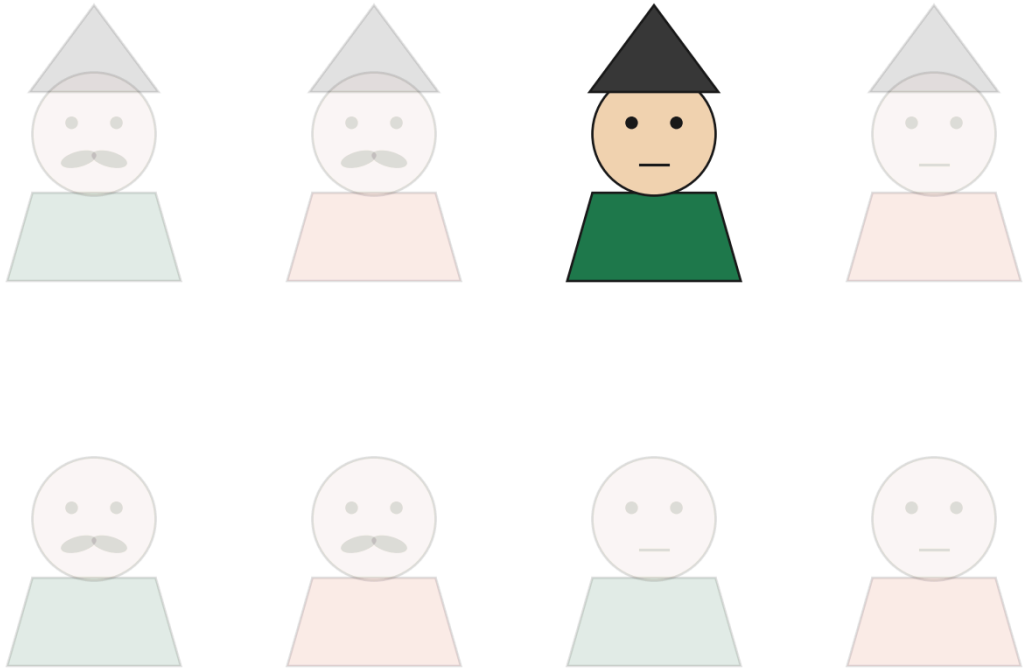
A witness saw one of them



Question	Answer	Bit	Left
			8
Wearing a hat?	yes	1	4
A mustache?	no	0	2

Eight Suspects

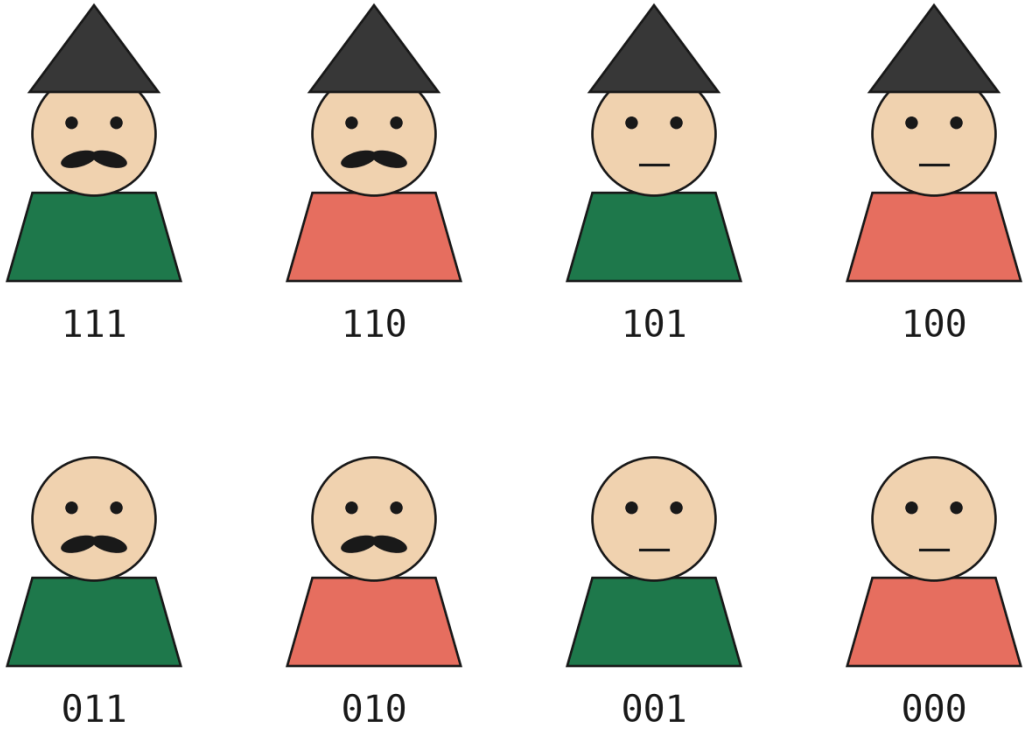
A witness saw one of them



Question	Answer	Bit	Left
			8
Wearing a hat?	yes	1	4
A mustache?	no	0	2
A green shirt?	yes	1	1

Three Bits for Eight Suspects

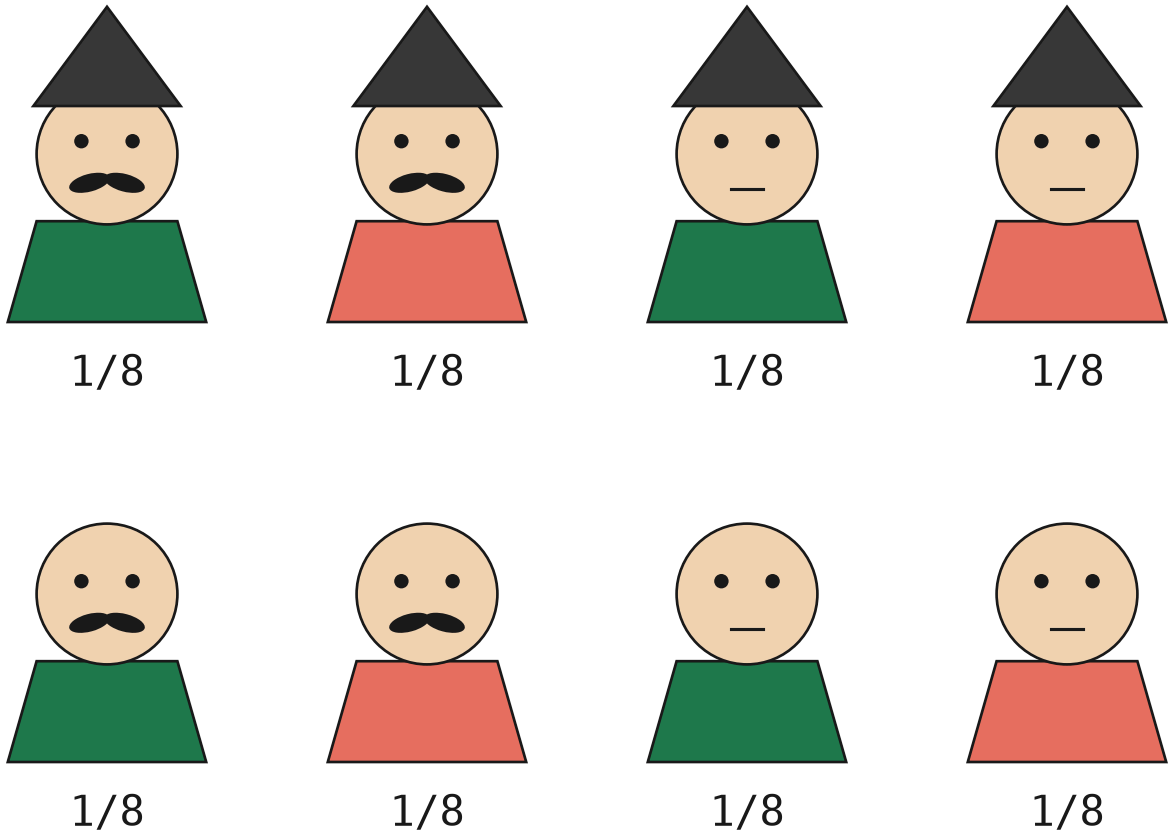
hat, mustache, green shirt: 1 for yes, 0 for no



Bits	Possibilities
1	2
2	4
3	8
10	1,024
20	1,048,576

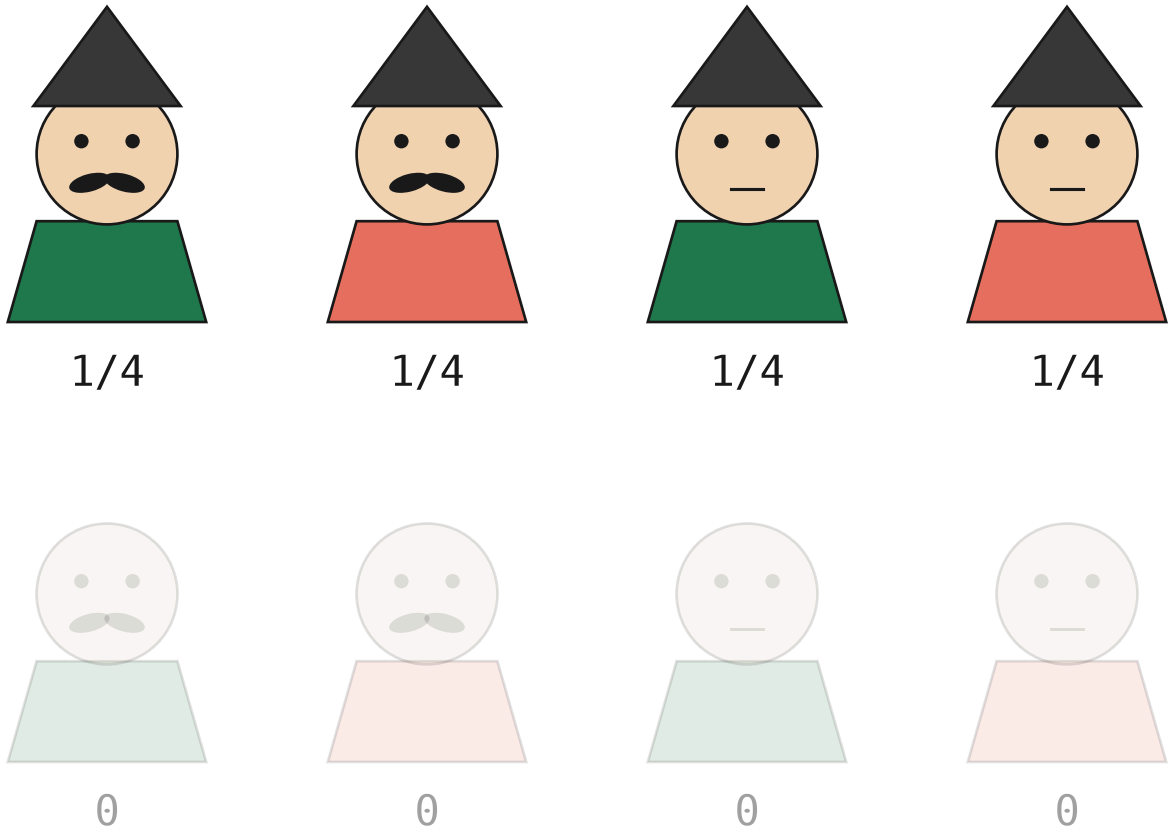
What an Answer Changes

Before any answer: each suspect is the one with probability 1/8



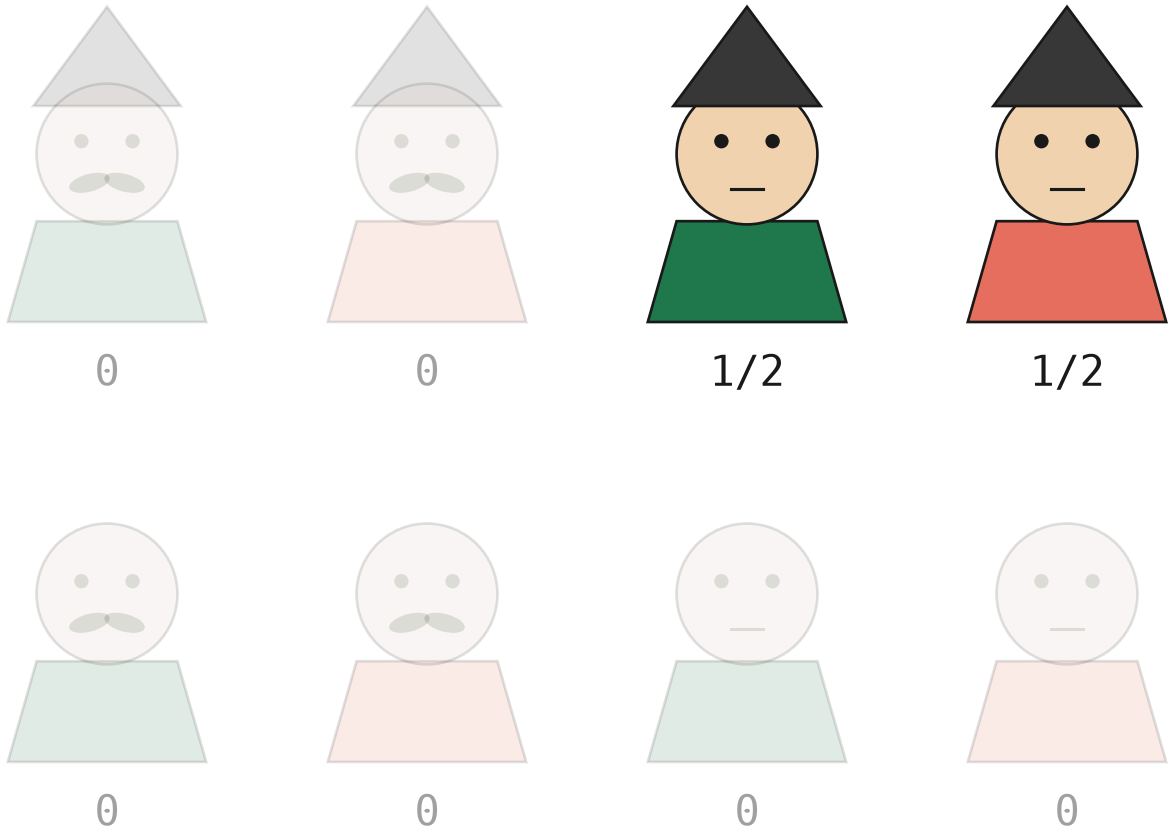
What an Answer Changes

"A hat": four drop to 0, and the other four rise to 1/4



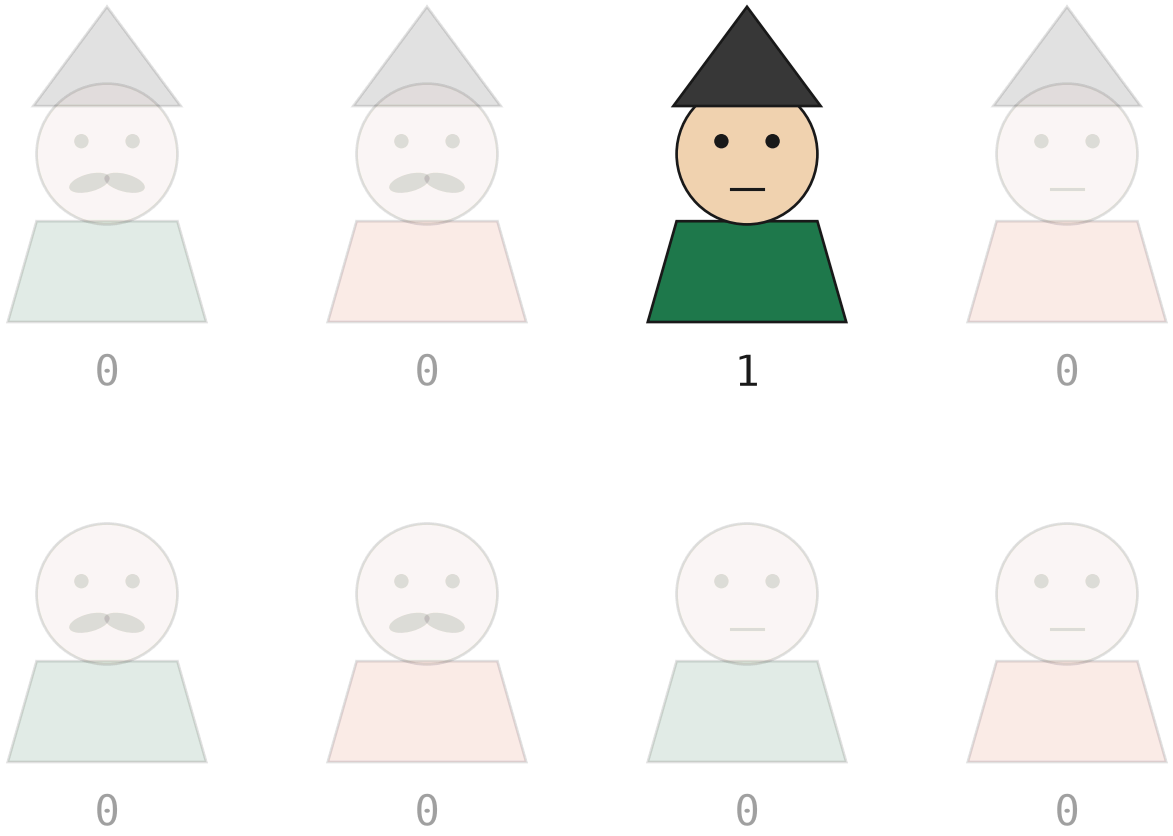
What an Answer Changes

"No mustache": two more drop to 0, and the other two rise to 1/2



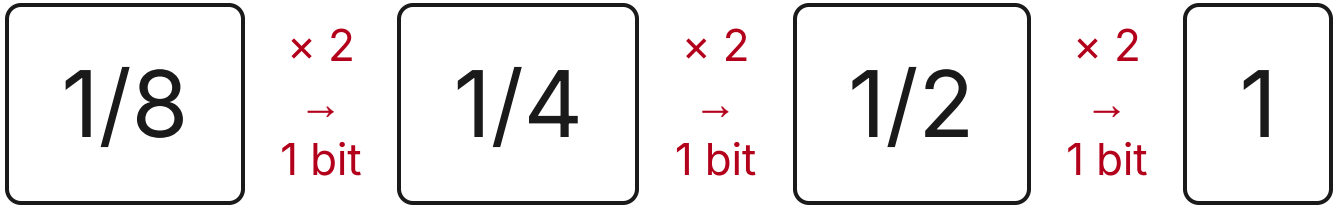
What an Answer Changes

"A green shirt": the true suspect reaches 1



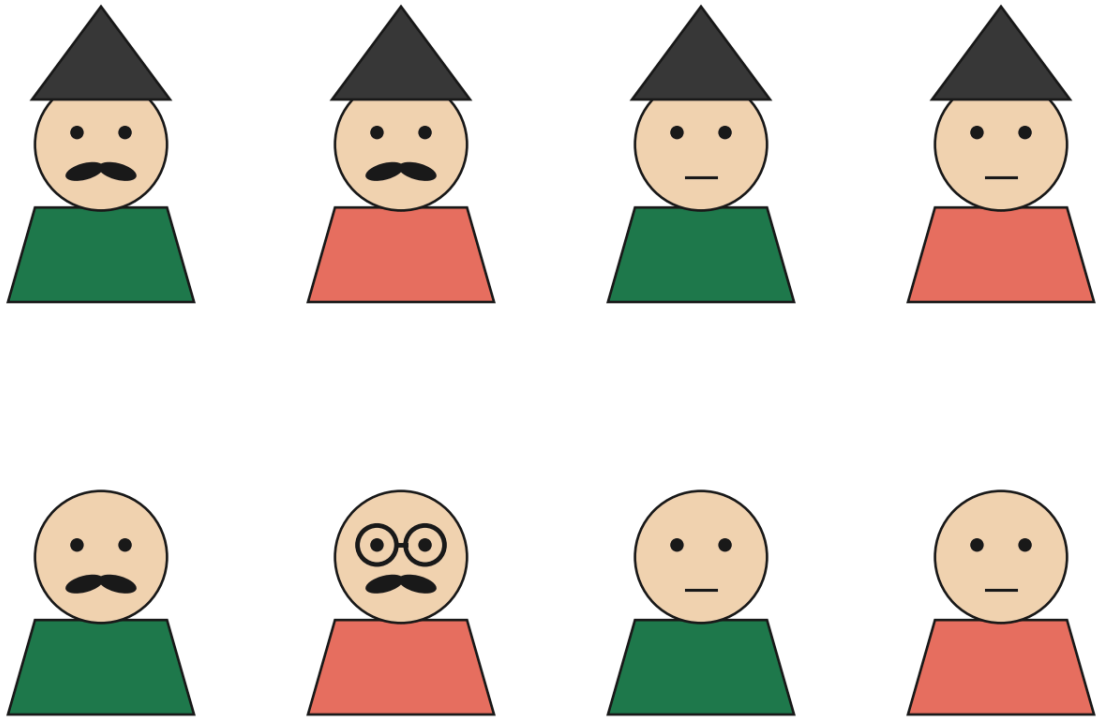
One Bit Doubles the Probability of the True Case

The probability that I give to the true suspect



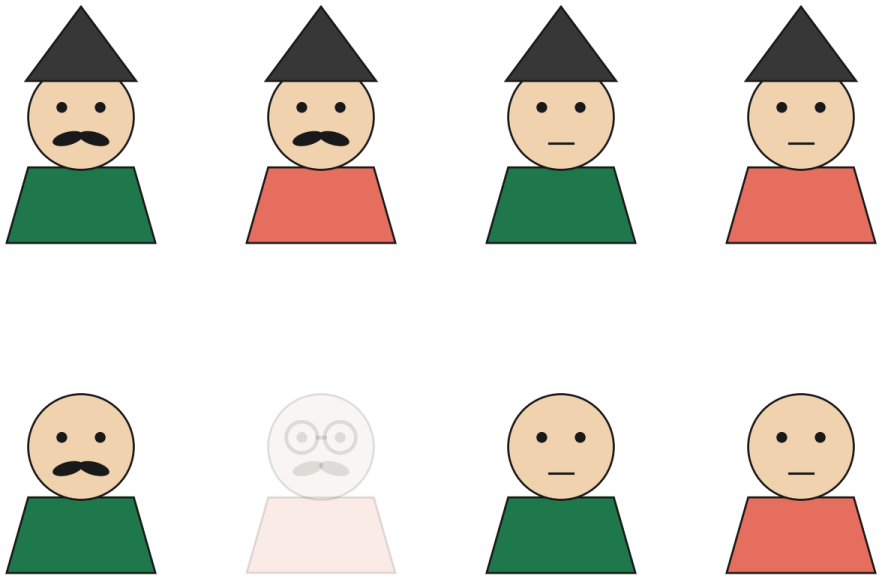
A Question That Does Not Split in Half

"Is the suspect wearing glasses?" Only one of the eight is.

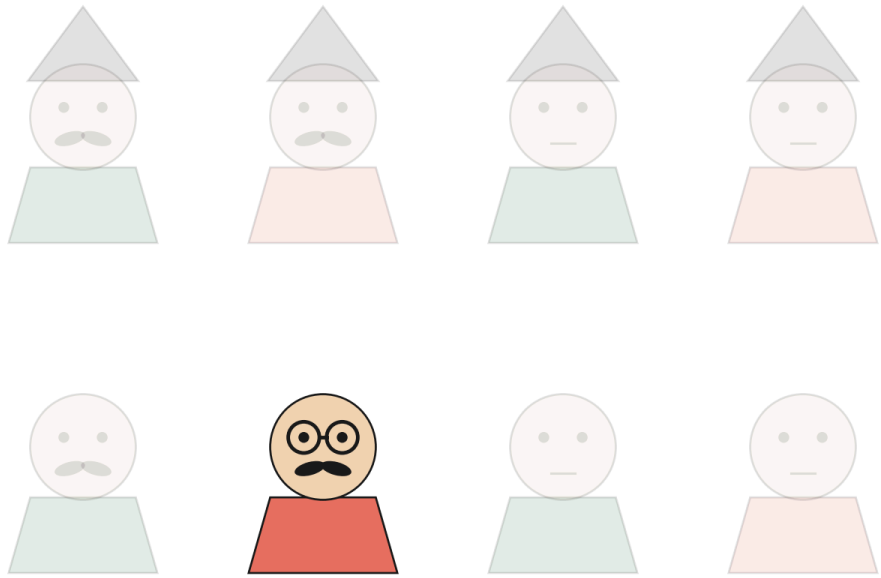


A Question That Does Not Split in Half

"Is the suspect wearing glasses?"

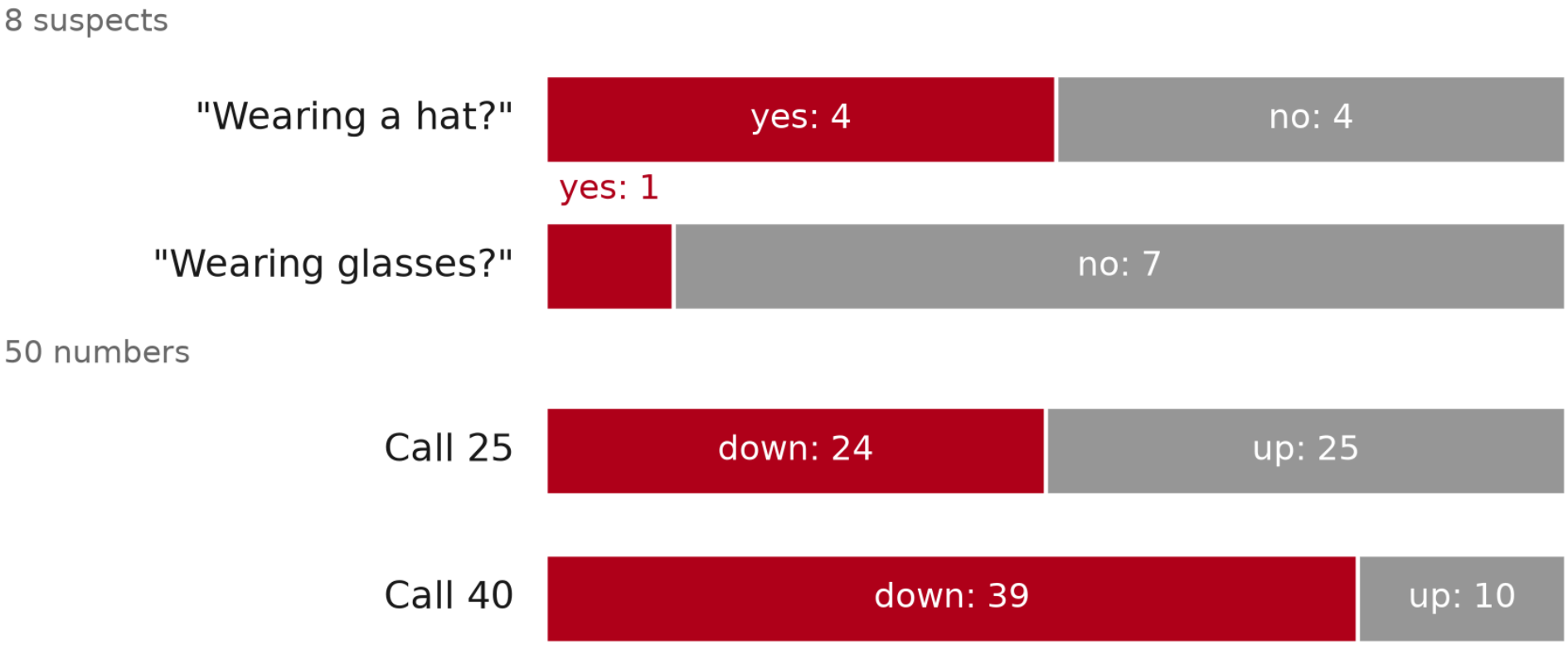


"no", 7 times in 8: 7 left



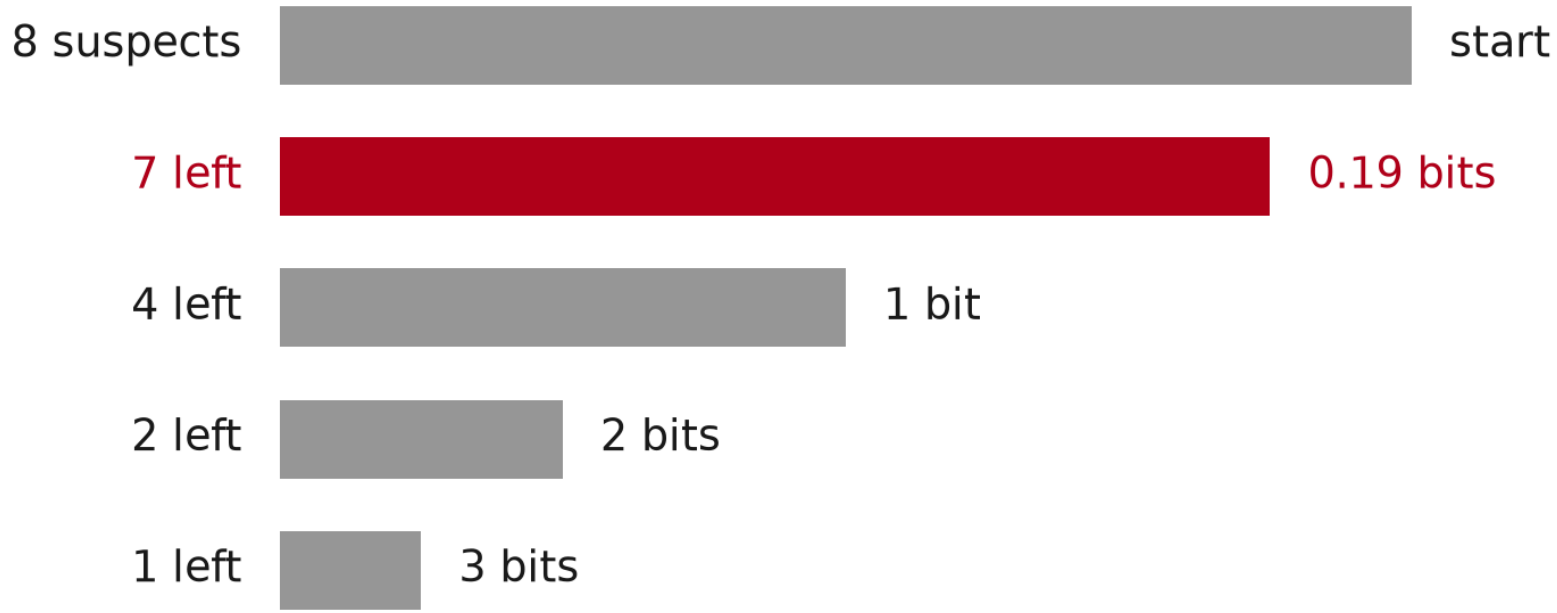
"yes", 1 time in 8: 1 left

Questions That Split in Half



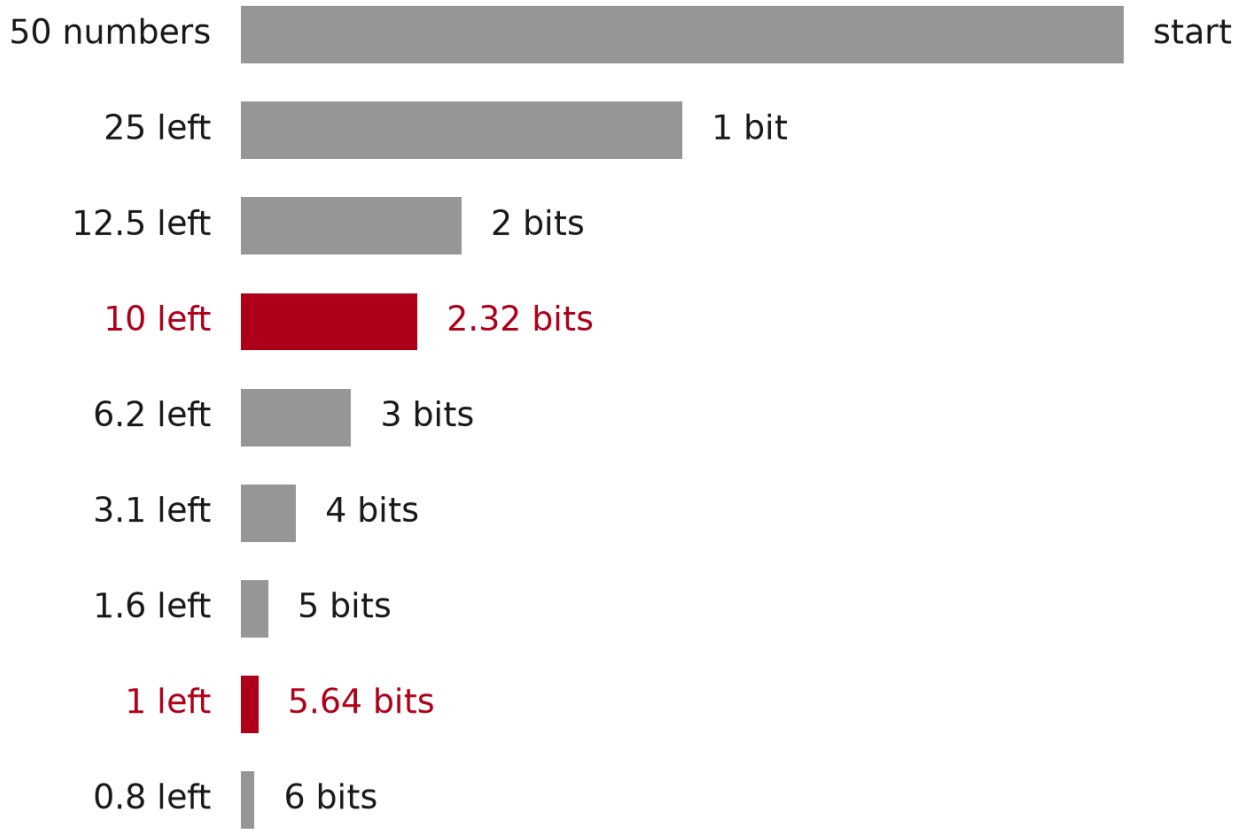
Counting Halvings

Information in an answer: how many times it halved the possibilities

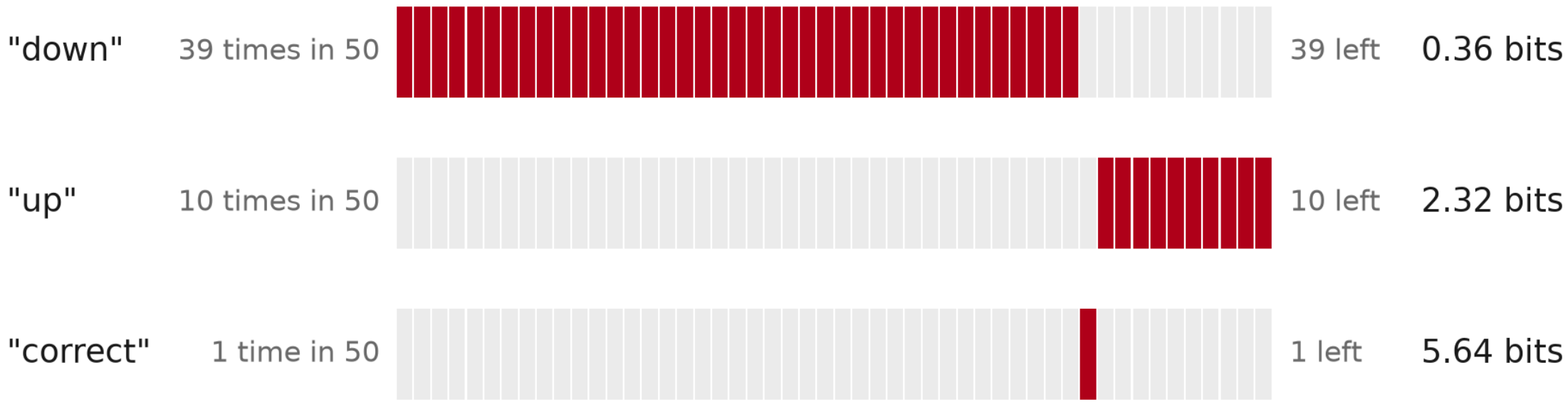


Counting Halvings

50 numbers

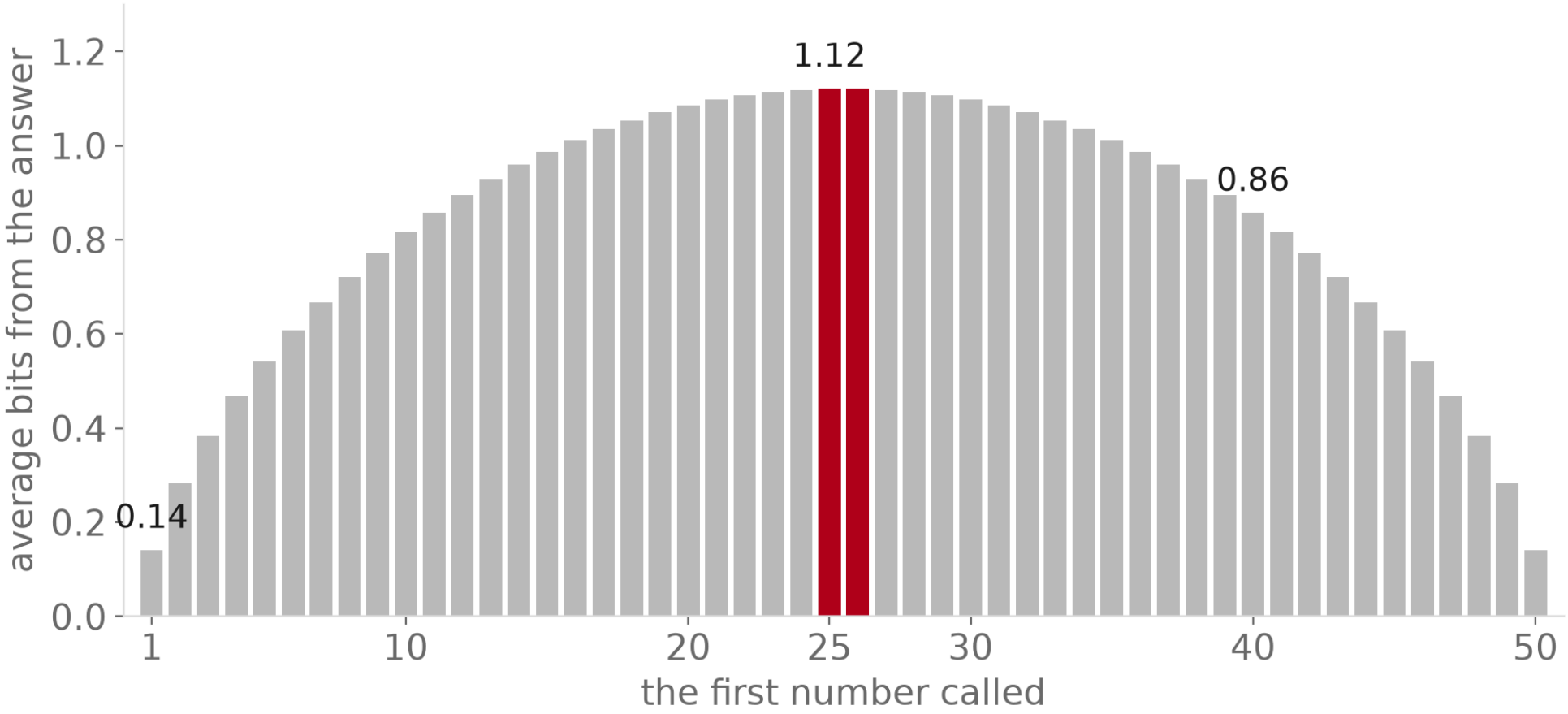


Calling 40 First



- Average: 0.86 bits

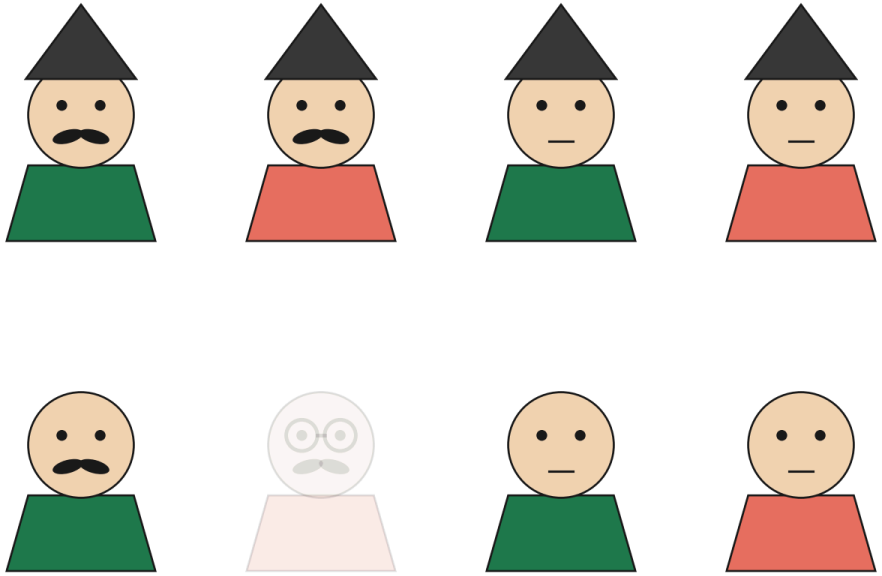
Average Information of the First Number



- Highest when "up" and "down" are equally likely

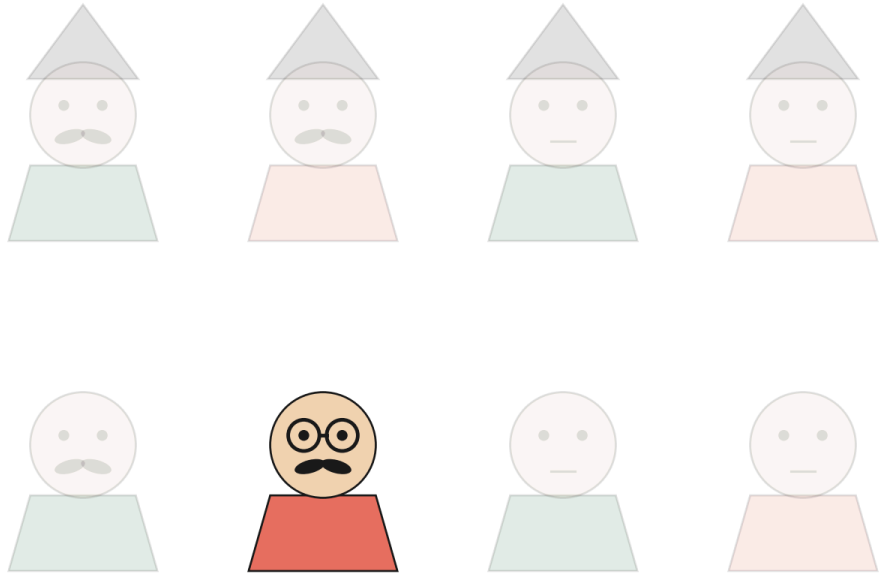
A Rare Answer Leaves Fewer Suspects

"Is the suspect wearing glasses?"



"no", 7 times in 8

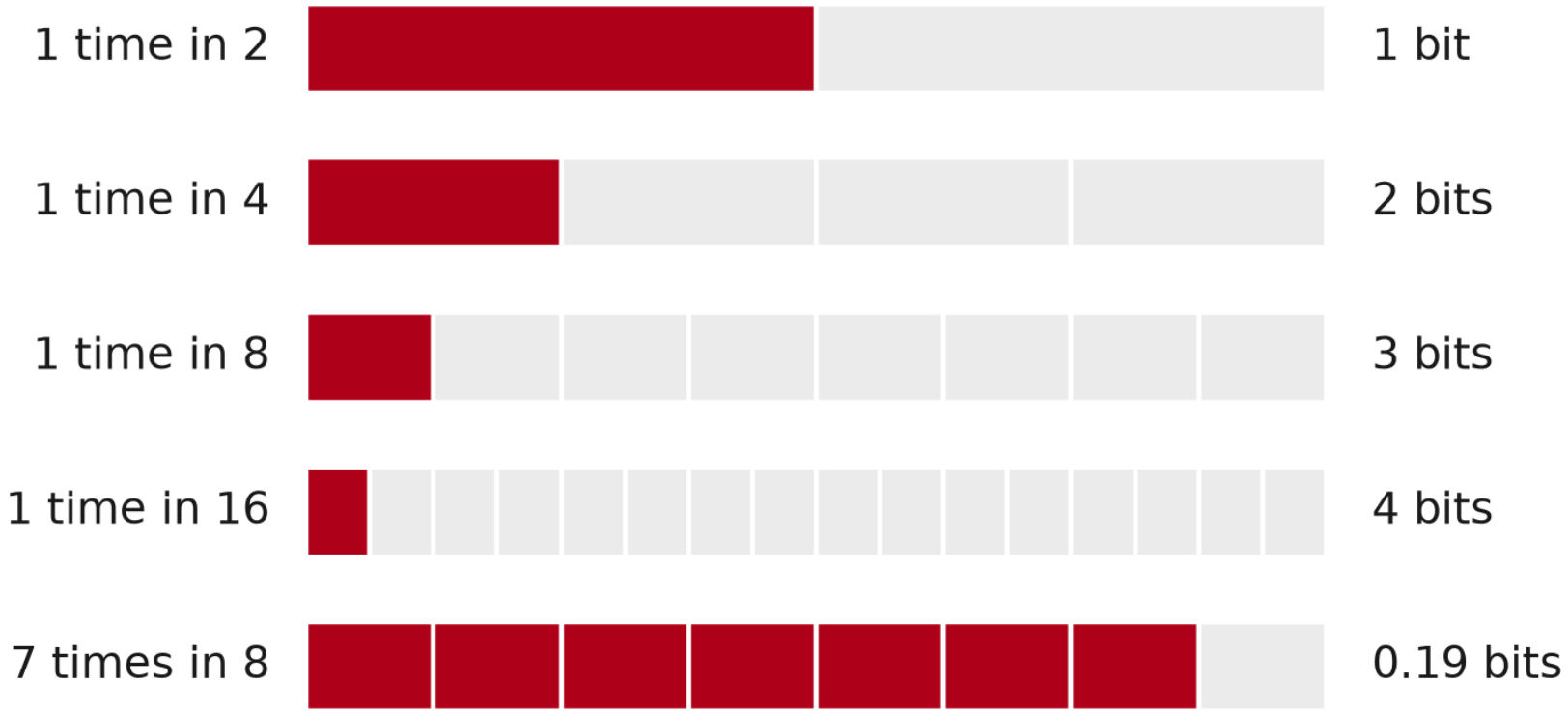
each of the 7: 1/8 → 1/7 (× 1.14), 0.19 bits



"yes", 1 time in 8

the one left: 1/8 → 1 (× 8), 3 bits

From "How Often" to Bits



- Each time the answer becomes half as likely, it carries 1 more bit

Information in One Piece of News

News that happens 1 time in N: as much as picking 1 out of N

News	How often	Information
A coin landed heads	1 in 2	1 bit
A die showed 6	1 in 6	2.58 bits
The number in the cap is 37	1 in 50	5.64 bits
This Lotto 6/45 ticket won first prize	1 in 8,145,060	22.96 bits
This Lotto 6/45 ticket did not win first prize	8,145,059 in 8,145,060	0.0000002 bits

- As a formula: information = $\log_2$ (1 / probability)

Words Carry Bits

The bits of each word, measured with a language model (Qwen2.5-7B)

Twinkle,	twinkle,	little	star,	how	I	wonder	what	you	are.
22.9	1.5	0.3	2.7	2.6	0.1	0.0	0.1	0.1	2.2

"Twinkle, Twinkle, Little Star"

To	be,	or	not	to	be,	that	is	the	question.
10.0	15.5	0.2	0.0	0.0	1.5	0.4	0.0	0.0	4.4

William Shakespeare, *Hamlet*, Act 3 Scene 1

Words Carry Bits

A song everyone knows

Happy	birthday	to	you.
15.3	3.2	1.4	7.1
Happy	birthday	to	you.
1.7	0.0	0.1	0.2
Happy	birthday	dear	Minji.
0.1	0.0	3.0	23.6
Happy	birthday	to	you.
0.1	0.0	0.0	1.1

Words Carry Bits

Two excuses

I'm	sorry	I'm	late.	The	subway	was	delayed.
11.0	4.0	11.1	8.3	3.7	7.5	0.8	6.3

I'm	sorry	I'm	late.	A	goose	was	chasing	me.
11.0	4.0	11.1	8.3	6.0	18.4	4.7	5.6	2.6

Words Carry Bits

A class notice

Assignment	3	is	due	on	Sunday	29	November	at	22:00.
16.1	3.7	9.2	1.9	1.5	4.4	8.9	3.8	2.2	10.2

The	one-page	proposal	is	due	on	Monday	23	November.
5.3	20.6	6.4	4.2	6.0	3.2	3.7	12.0	7.4

Proposal	presentations	are	on	Tuesday	24	November	and	Thursday	26	November.
10.1	5.6	2.5	1.9	3.0	0.9	0.2	3.5	3.7	0.1	1.4

Words Carry Bits

One novel: the famous first sentence, and a sentence from the middle

It	is	a	truth	universally	acknowledged,	that	a	single	man	in	possession
9.0	1.5	3.3	7.5	0.1	0.4	0.0	0.1	0.4	0.2	0.0	0.0

of	a	good	fortune	must	be	in	want	of	a	wife.
0.0	0.0	0.0	0.0	2.8	0.0	0.0	0.0	0.0	0.0	1.4

Jane Austen, *Pride and Prejudice* (1813), Chapter 1, first sentence: 26.8 bits in 23 words

She	was	sorry	that	Mr.	Darcy	should	have	delivered	his
12.9	3.3	12.1	2.1	8.2	6.0	3.4	0.9	13.9	3.6

sentiments	in	a	manner	so	little	suited	to	recommend	them.
2.8	0.8	5.0	0.3	0.2	1.7	4.1	0.1	10.7	2.9

Pride and Prejudice, Chapter 40: 95.1 bits in 20 words

Twenty Questions

- 20 yes-or-no answers, each leaving half: 1 in 1,048,576
- A birthday: 1 in 365, about 9 questions

Entropy

The average information of one answer, before you hear it

Source	Entropy
A coin that always lands heads	0 bits
A coin that lands heads 9 times in 10	0.47 bits
A fair coin	1 bit
A fair die	2.58 bits
A fair draw from 1 to 50	5.64 bits